

Python Placement Training - Day 2 Notes

List Working over Array

Lists store object references instead of actual values directly.

Address Formula

Address = Base Address + (Offset × Size of Data Type)

Integer List Input

```
l = list(map(int, input().split()))
```

String List Input

```
l = input().split()
```

append()

Adds an element at the end of the list.

insert()

Adds an element at a specific position.

extend()

Adds iterable elements to the list.

append() vs extend()

append() adds single item.
extend() works only with iterable items.
extend(5) gives TypeError.

pop()

Removes elements from list.
Without index, removes last element.

clear()

Removes all elements from the list.

sort()

Sorts list in ascending order.
reverse=True gives descending order.

reverse()

Reverses list elements.

index()

Returns index position of element.

count()

Returns frequency of an element.

copy()

Creates a separate copy of list.

Tricky remove() Question

Removing elements while iterating changes indexes and skips elements.

Example

```
l1 = [1,2,3,4,5,6,7]
for i in l1:
    l1.remove(i)
Output: [2, 4, 6]
```

List Slicing

Syntax: list[start:end:step]

Built-in Functions

Functions like print(), len(), type() are used without dot operator.

Concatenation

Used to join sequences using +

Repetition

Used to repeat sequences using *

Membership Operators

in and not in check presence of elements.

Removing Duplicates

Use if i not in l1 to avoid duplicate values.